

Square Numbers & Their Properties

Read the key idea, learn the traps that cost marks, then study the worked examples — each one is written out line by line, exactly the way you should write it in the exam.

What is a square number?

The idea: When a number is multiplied by **itself**, the result is a **square number** (also called a **perfect square**). We write it with a small raised 2.

$$5 \times 5 = 5^2 = 25 \rightarrow \text{read as "5 squared"}$$

Why "square"? A square of side 5 contains exactly $5 \times 5 = 25$ unit squares. The area of a square *is* the square of its side.

Side	Working	Square (area)
1	1×1	1
2	2×2	4
3	3×3	9
4	4×4	16
5	5×5	25
10	10×10	100

Memorise the first 20 perfect squares — they appear again and again in MCQs:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100,
121, 144, 169, 196, 225, 256, 289, 324, 361, 400

The 4 properties that get tested

- Units-digit rule.** A perfect square can only end in **0, 1, 4, 5, 6, or 9** — *never* in **2, 3, 7, or 8**.
- Trailing-zeros rule.** A perfect square always has an **even** number of zeros at the end ($100 \rightarrow 2$ zeros, $10000 \rightarrow 4$). So 1000 (3 zeros) is not a square.
- Parity.** The square of an **even** number is even; the square of an **odd** number is odd. ($6^2=36$, $7^2=49$.)
- Sum of odd numbers.** $1=1^2$, $1+3=4=2^2$, $1+3+5=9=3^2$, $1+3+5+7=16=4^2$. The sum of the first **n** odd numbers is **n^2** .

Two bonus facts examiners love:

- Count of numbers *between* n^2 and $(n+1)^2 = 2n$. (Between 25 and 36: $2 \times 5 = 10$.)
- Gap between consecutive squares n^2 and $(n+1)^2 = 2n + 1$.

Prerequisite refresher: "units digit" = the last digit; "parity" = whether a number is odd or even; "consecutive odd numbers" = 1, 3, 5, 7, ... with no gaps.

Where marks slip away

Trap 1 — Reversing the units-digit rule. Ending in 0,1,4,5,6,9 does **not** prove a number is a square. 26 ends in 6 but is not a square. Use the rule only to rule numbers *out*.

Trap 2 — "Square" $\neq$ "double". $5^2 = 25$, not 10. Squaring is $\times$ itself.

Trap 3 — "How many between". Numbers strictly between n^2 and $(n+1)^2 = 2n$ (not $2n+1$, not n).

Trap 4 — Trailing zeros. Always count the zeros. An odd number of zeros (e.g. 1000) means it cannot be a square.

Slip 5 — Notation. Write the index small and raised: 7^2 . Writing "72" loses the meaning and the mark.

Model answers — write them like this

Example A. Is 24 a perfect square?

Step 1: Check the units digit. 24 ends in 4, which is an allowed digit, so it is not ruled out yet.

Step 2: List nearby squares: $4^2 = 16$ and $5^2 = 25$. So 24 lies between two consecutive squares.

Step 3: Since 24 is not equal to any whole number squared, it is not a perfect square.

$\therefore$ 24 is NOT a perfect square

Example B. How many natural numbers lie between 5^2 and 6^2 ?

Step 1: Use the rule: numbers between n^2 and $(n+1)^2 = 2n$.

Step 2: Here $n = 5$, so the count = $2 \times 5 = 10$.

Check: $5^2 = 25$ and $6^2 = 36$. The numbers 26, 27, ..., 35 number exactly 10.

$\therefore$ 10 numbers

Example C. Find the units digit of the square of 162.

Step 1: Only the units digit of the original number affects the units digit of the square.

Step 2: The units digit of 162 is 2, and $2^2 = 4$.

Step 3: So the square of 162 ends in 4.

$\therefore$ Units digit = 4

Example D. Show that 36 is a perfect square by repeated subtraction of odd numbers, and find its square root.

Method: Subtract the odd numbers 1, 3, 5, 7, ... one at a time until the result is 0.

Working: $36 - 1 = 35$; $35 - 3 = 32$; $32 - 5 = 27$; $27 - 7 = 20$; $20 - 9 = 11$; $11 - 11 = 0$.

Count: We subtracted 6 odd numbers to reach exactly 0.

Conclude: Reaching 0 means 36 is a perfect square, and the count (6) is its square root.

$\therefore$ 36 is a perfect square and $\sqrt{36} = 6$

Example E. Given $35^2 = 1225$, find 36^2 using the odd-number pattern.

Step 1: The gap between n^2 and $(n+1)^2$ is $(n) + (n+1)$. Here that is $35 + 36 = 71$.

Step 2: So $36^2 = 35^2 + 71 = 1225 + 71$.

Step 3: $1225 + 71 = 1296$.

$\therefore 36^2 = 1296$

Next: [try the practice worksheet](#), [then check the answer key](#).